

Pearls AQI Predictor: Historical Data Backfill and Feature Store Population

Milestone 3 Technical Report

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Abstract—This report documents the historical data backfill phase of the Pearls AQI Predictor, a serverless machine learning system for forecasting the Air Quality Index (AQI) of Karachi. The objective of this milestone was to populate the Hopsworks Feature Store with approximately six months of hourly historical data to support subsequent exploratory analysis and model training. A backfill script was developed to fetch historical weather and pollution data and engineer the required features. During this process, a data-quality limitation in the initially selected pollution source was identified and resolved by adopting the Open-Meteo Air Quality API, from which AQI was computed using the United States Environmental Protection Agency (US EPA) breakpoint formula. The final dataset comprises 4,344 hourly records with 18 features, successfully materialized in the Feature Store.

Index Terms—historical backfill, feature store, Hopsworks, Open-Meteo, AQI, data pipeline

I. INTRODUCTION

Machine learning models for time-series forecasting require a substantial volume of historical data to learn meaningful patterns. While the feature pipeline developed in the previous milestone captures live data on an hourly basis, this alone would take months to accumulate enough records for training. The purpose of the backfill milestone is therefore to retroactively populate the Hopsworks Feature Store with historical data, providing an immediate foundation for analysis and modeling.

This report describes the design of the backfill process, the data-quality challenge encountered, the corrective action taken, and the successful population of the Feature Store.

II. BACKFILL DESIGN

A dedicated backfill script was developed to retrieve approximately six months of hourly historical data for Karachi. The script performs three main stages: data retrieval from external APIs, feature engineering, and insertion into the Hopsworks Feature Store.

The feature set mirrors that of the live feature pipeline to ensure schema consistency. It includes pollutant concentrations, meteorological variables, time-based attributes (hour, day of week, month, weekend indicator), the computed AQI value, and a categorical AQI health class. The primary key is composed of the city and timestamp, with the timestamp serving as the event time.

III. DATA-QUALITY CHALLENGE AND RESOLUTION

The initial implementation retrieved pollution data from the AQICN API. However, analysis of the resulting dataset revealed that all historical records carried an identical AQI value. Investigation determined that the AQICN free tier returns only the *current* air quality reading rather than true historical observations. Consequently, the same present-day value was stamped onto every historical timestamp, producing a constant AQI series with zero variance.

A constant target variable is unsuitable for predictive modeling, as there is no variation for a model to learn. To resolve this, the pollution data source was changed to the Open-Meteo Air Quality API, which provides genuine historical hourly concentrations of PM2.5, PM10, NO₂, O₃, SO₂, and CO. The AQI was then computed from the PM2.5 concentration using the US EPA breakpoint formula:

$$AQI = \frac{I_{hi} - I_{lo}}{C_{hi} - C_{lo}}(C - C_{lo}) + I_{lo} \quad (1)$$

where C is the measured PM2.5 concentration, C_{lo} and C_{hi} are the surrounding concentration breakpoints, and I_{lo} and I_{hi} are the corresponding AQI breakpoints. Weather data continued to be sourced from the Open-Meteo Archive API. This correction produced a realistic AQI series ranging from 15 to 172.

IV. FEATURE STORE POPULATION

The engineered dataset was inserted into the Hopsworks Feature Store as the feature group `aqi_features`. During materialization, it was observed that the offline materialization job on the Hopsworks free tier was significantly slower than expected and, in some cases, failed to persist the full dataset. This was resolved by recreating the feature group cleanly and inserting the data while explicitly waiting for the materialization job to complete. After this adjustment, the materialization succeeded and a read-back confirmed the full dataset of 4,344 records with 18 features and 4,344 unique timestamps.

V. CONCLUSION

The historical backfill milestone successfully populated the Hopsworks Feature Store with six months of hourly data for Karachi. A significant data-quality limitation in the initial pollution source was identified and resolved by adopting the

TABLE I
FINAL BACKFILL DATASET SUMMARY

Attribute	Value
Total records	4,344
Number of features	18
Time span	Approximately 6 months
Temporal resolution	Hourly
AQI range	15 to 172
Storage	Hopsworks Feature Store

Open-Meteo Air Quality API together with the US EPA AQI computation, yielding a realistic and varied dataset. Challenges with free-tier materialization were also overcome through a clean re-insertion strategy. The resulting dataset of 4,344 records provides a robust foundation for the exploratory data analysis and model training conducted in the subsequent milestones.